

Atomic Physics

Report Assignment

Instructor: Munekazu Horikoshi

Due date: Monday, July 27, 2026

Submission: Moodle

Instructions

- Answer all questions in English.
 - Show the key steps of your calculations where appropriate.
 - Clearly explain your reasoning.
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Problem 1: Quantization of the Electromagnetic Field and Atom-Field Interactions

Answer the following questions.

1. Explain why the transition probability for photon absorption is proportional to the photon number n .
2. Explain why the transition probability for photon emission is proportional to $n + 1$.
3. In the factor $n + 1$, explain what the terms proportional to n and to 1 physically represent.
4. Explain the difference between spontaneous emission and stimulated emission, and discuss why spontaneous emission can occur even in vacuum, relating your answer to the role of the quantized electromagnetic field.
5. The lifetime of the 2^3S_1 state of helium is about 8000 s, whereas typical electric-dipole-allowed transitions have lifetimes of the order of 10^{-8} s. Using the expression for the spontaneous emission rate derived in class, explain the physical origin of this enormous difference.

Problem 2: Thermal Equilibrium of Atoms in Blackbody Radiation

An atom interacting with blackbody radiation undergoes

- absorption,
- stimulated emission,
- spontaneous emission.

Answer the following questions.

1. Explain why the internal-state populations of atoms interacting with blackbody radiation obey the

Boltzmann distribution in thermal equilibrium. In your explanation, refer to the balance between absorption, stimulated emission, and spontaneous emission.

2. Explain how absorption, stimulated emission, and spontaneous emission contribute to
 - the friction force acting on the atom, and
 - momentum diffusion.

Problem 3: Thermalization in Blackbody Radiation

Consider sodium atoms with a resonance wavelength

$$\lambda = 589 \text{ nm},$$

interacting with blackbody radiation at the temperature of the solar surface,

$$T = 6000 \text{ K}.$$

The first ionization energy of sodium corresponds to a temperature of approximately

$$6.0 \times 10^4 \text{ K},$$

so ionization can be neglected.

Answer the following questions.

1. Calculate the average photon number $\bar{n}(\omega_A)$ at the resonance frequency.
2. Using the expression for the friction coefficient, or equivalently the thermalization time, derived in class, estimate the thermalization time of sodium atoms.
3. Based on your results, explain why blackbody radiation alone cannot efficiently cool atoms.